



Dayananda Sagar College of Engineering
(An Autonomous Institute affiliated to VTU, Approved by AICTE & ISO 9001:2008 Certified)
Accredited by National Assessment & Accreditation Council (NAAC) with 'A' grade
Shavige Malleshwara Hills, Kumaraswamy Layout, Bengaluru-560078.



Department of Master of Computer Applications

A Project Report On
**“CropXpert: AI-Powered Precision
Agriculture with Integrated Sustainability
Intelligence”**

Submitted by

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1DS24MC075

Under the guidance of

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CERTIFICATE

This is to certify that the Dissertation work entitled "CropXpert : AI-Powered Precision Agriculture with Integrated Sustainability Intelligence" carried out by Pranav Ramachandra Hegde (1DS24MC075) in partial fulfillment of the award of the Degree of Master of Computer Applications of Dayananda Sagar College of Engineering, an Autonomous Institute affiliated to VTU, Belagavi, during the year 2025-2026. It is certified that all suggestions or corrections indicated for internal assessment have been incorporated into the report.

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DECLARATION

I, Pranav Ramachandra Hegde, student of III Semester MCA, Dayananda Sagar College of Engineering, USN 1DS24MC075, hereby declare that the project entitled "CropXpert: AI-Powered Precision Agriculture with Integrated Sustainability Intelligence" was completed by me under Dr.V. Srinivasan, Associate Professor, Department of MCA, DSCE, Bangalore. This project is submitted in partial fulfillment of the requirements for the award of the degree of Master of Computer Applications by Visvesvaraya Technological University during the academic year 2025.

Pranav Ramachandra Hegde
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I dedicate this page to those who are responsible for bringing a proper shape to this project. Without their guidance and help, this project would not have been so smooth and efficient.

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ABSTRACT

Agriculture remains the most significant sector of the global economy, particularly in developing nations where it serves as the primary livelihood for a vast majority of the population. However, the sector is increasingly vulnerable to fluctuating climatic conditions and declining soil productivity. Traditional agricultural practices, while time-tested, often fail to optimize crop selection in the face of modern environmental stressors. This project addresses the need for a scientific, data-driven approach to farming through the development of a comprehensive Crop Recommendation System.

The **Crop Recommendation System** is a sophisticated web application that integrates advanced machine learning techniques with environmental sustainability metrics. At its core, the system utilizes a Random Forest Classifier to process seven critical soil and climatic parameters: Nitrogen (N), Phosphorus (P), Potassium (K), Temperature, Humidity, pH value, and Rainfall. By analyzing historical agricultural datasets, the system identifies the most suitable crop for a specific plot of land, thereby maximizing potential yield and minimizing the risks associated with crop failure.

Beyond productivity, this project incorporates a strong emphasis on environmental stewardship through a dedicated **Carbon Footprint Calculator**. This module allows users to quantify the carbon emissions generated by farming operations, specifically focusing on fertilizer application, fuel for machinery, and electricity for irrigation. By providing farmers with a clear understanding of their environmental impact, the system encourages the adoption of sustainable farming practices and promotes the long-term health of the ecosystem.

The technical architecture of the system is designed for high performance and scalability. The frontend is developed using **React.js**, ensuring a responsive user experience across desktop and mobile devices. The backend leverages a microservices approach, with **Node.js** managing user authentication and application state, while multiple **Python (Flask)** services handle the execution of machine learning models. Data persistence is managed via **MongoDB**, allowing for flexible and efficient storage of user history and prediction data.

Chapter 1

INTRODUCTION

Agriculture has been the bedrock of human civilization since the Neolithic Revolution, providing the essential nutrients required for survival and the raw materials necessary for the development of industry. In the modern era, the significance of agriculture has only intensified, as the global population approaches 8 billion and the demand for food security becomes a paramount concern for nations worldwide. In the context of the Indian economy, agriculture occupies a position of uniquely critical importance. It is not merely a sector of the economy but a way of life for millions, employing approximately 58% of the population and contributing significantly to the Gross Domestic Product (GDP).

Despite its historical and economic magnitude, the agricultural sector in the 21st century is facing an unprecedented convergence of challenges. Climate change has introduced a level of volatility in weather patterns that traditional farming practices are ill-equipped to handle. Rising global temperatures, erratic rainfall distributions, and the increasing frequency of extreme weather events such as droughts and floods have made agricultural output increasingly unpredictable. Furthermore, decades of intensive farming characterized by the overuse of chemical fertilizers and the neglect of soil health have led to widespread land degradation, reducing the natural productivity of the soil.

The concept of "What to plant?"—once governed by centuries of inherited wisdom and seasonal predictability—has now become a high-stakes gamble for the modern farmer. Traditionally, crop selection decisions were based on local trends, historical family practices, or a rudimentary understanding of the current market prices. However, such intuitive methods often fail to account for the complex, non-linear interactions between soil chemistry, local micro-climates, and the biological requirements of specific crop varieties.

The bridge between advanced scientific agricultural research and the individual farmer in the field remains dangerously thin. While research institutions develop sophisticated models for soil health and crop resilience, this knowledge often fails to reach the person behind the plow in a format that is accessible, timely, and actionable. Precision Agriculture (PA) aims to close this gap. By utilizing information technology including Global Positioning Systems (GPS), the Internet of Things (IoT), and Machine Learning (ML) Precision Agriculture enables farmers to observe, measure, and respond to variability in crops and soil at a granular level.

The **Crop Recommendation System** proposed in this project represents a significant contribution to the field of Precision Agriculture. It is designed to empower farmers with data-driven insights, replacing intuition with scientific evidence. By processing key environmental and soil parameters through a robust machine learning algorithm, the system provides a personalized recommendation for the most suitable crop for a specific plot of land. This project goes a step further by integrating environmental sustainability into the decision-making process, incorporating a Carbon Footprint Calculator to help farmers understand the environmental impact of their choices.

The Critical Role of Soil and Environmental Parameters:

The success of any agricultural venture is fundamentally determined by the interaction between the genetic potential of the crop and the environmental conditions in which it is grown. The Crop Recommendation System focuses on seven primary parameters that define this environment:

Nitrogen (N): As a primary macronutrient, nitrogen is essential for leaf growth and the development of the plant's green color (chlorophyll). It is a key component of amino acids and proteins. Nitrogen deficiency often results in stunted growth and yellowing of leaves.

Phosphorus (P): Crucial for root development and the production of seeds and fruit. It plays a vital role in energy transfer within the plant. Phosphorus is particularly important during the early stages of plant growth.

Potassium (K): Helps regulate the plant's water balance and is essential for the activation of enzymes. It improves the plant's resistance to diseases and pests and enhances the quality of the harvested produce.

Temperature: Every crop has an optimal temperature range for germination, growth, and reproduction. Deviations from this range can stunt development, reduce fertility, or lead to complete crop failure in extreme cases.

Humidity: Affects the rate of transpiration and the plant's susceptibility to certain fungal and bacterial diseases. High humidity combined with specific temperatures often creates a breeding ground for pests.

pH Value: Measures the acidity or alkalinity of the soil. Many nutrients become chemically unavailable to plants if the pH is too high (alkaline) or too low (acidic). Most crops thrive in a slightly acidic to neutral range.

Rainfall: Provides the necessary water for photosynthesis and nutrient uptake. Both excessive rainfall (leading to waterlogging) and insufficient rainfall (leading to drought stress) can be detrimental to crop health.

Detailed Problem Statement:

In the current agricultural landscape, the lack of data-driven decision-making tools leads to several systemic inefficiencies that threaten the stability of the rural economy:

Suboptimal Yields and Economic Waste: Due to incorrect crop selection, farmers often achieve yields far below the land's potential capacity. This represents a massive waste of land, labor, and capital resources on a national scale.

The Cycle of Rural Debt: Small-scale farmers are particularly vulnerable to the financial impact of a single failed crop season. The loss of investment in seeds and fertilizers often leads to long-term debt, which can have devastating social consequences.

Environmental Degradation through Over-fertilization: The generic application of fertilizers—regardless of the soil's actual nutrient levels—results in nutrient runoff into water bodies, groundwater contamination, and soil acidification.

Vulnerability to Climate Insecurity: Farmers lack the analytical tools to adapt their planting strategies to the shifting climatic zones caused by global warming. A crop that was suitable ten years ago may no longer be viable today.

Lack of Sustainability Awareness: While agriculture is a major contributor to global greenhouse gas emissions, farmers have no mechanism to measure or mitigate their individual carbon footprint, hindering the transition to "Green Agriculture."

Project Objectives

The primary objective of this project is to create an integrated ecosystem for smart and sustainable farming. The specific goals are:

To Build a High-Precision Prediction Engine: Develop a Machine Learning model using the Random Forest algorithm that can categorize 22 different crop types based on soil and weather data with an accuracy exceeding 99%.

To Quantify and Mitigate Environmental Impact: Implement a Carbon Footprint Calculator that translates agricultural inputs (fertilizers, fuels, electricity) into CO_2 equivalents ($kgCO_2e$), providing a detailed sustainability report to promote eco-friendly practices.

To Ensure Universal Digital Accessibility: Design a responsive web interface using the MERN stack that provides a seamless experience across all digital devices, with a focus on mobile-first design for rural areas.

To Develop a Scalable Microservices Architecture: Ensure the computational intensity of the ML model does not impact the responsiveness of the user interface by decoupling the services using a Flask-based backend for analytics and Node.js for management.

To Promote Knowledge Transfer and Support: Include educational modules such as an AI-powered Chatbot (Agri-Mitra) and an Information Portal to provide real-time assistance and agricultural best practices.

Scope of the Project

The scope of the Crop Recommendation System extends beyond simple prediction to provide a holistic agricultural management tool:

For the Individual Farmer: Provides a personal dashboard to track soil health, prediction history, and environmental impact over multiple seasons.

For Agricultural Extension Officers: Serves as a digital assistant to provide data-backed advice during field visits, improving the quality of extension services.

For the Agri-Input Value Chain: Includes an integrated store module allowing farmers to purchase the specific fertilizers and seeds recommended by the system, reducing the purchase of unnecessary inputs.

For Researchers and Policy Makers: Provides a framework for regional soil health mapping and sustainability monitoring, which could eventually inform national agricultural strategies

.

Methodology: The Spiral Model Approach

This project adopts the **Spiral Model** of software development, which is particularly suited for projects involving complex algorithms and evolving user requirements. The spiral model allows for the iterative refinement of the system through successive cycles of planning, risk analysis, engineering, and evaluation. This iterative nature is crucial for agricultural projects where the feedback loop between the analytical model and the user interface design must be tightly coupled to ensure the final recommendation is both accurate and understandable.

Planning Phase:

In this initial stage, the project requirements were gathered through literature reviews and stakeholder analysis. The dataset selection was finalized, focusing on the validated Kaggle Crop Recommendation dataset containing 2200 records. This phase also involved defining the project's milestones, such as the delivery of the core ML engine and the subsequent integration of the sustainability modules.

Risk Analysis Phase:

Critical risks such as model bias, dataset imbalance, and architectural latency were identified. Mitigation strategies, such as using the Random Forest algorithm for its robustness against overfitting and a microservices architecture for performance, were developed. Each risk was quantified based on its potential impact on the farmer's decision-making process.

Engineering and Implementation Phase:

This involved the construction of the system. First, the ML model was trained and validated in Python using metrics like accuracy, precision, and recall. Next, the Node.js backend and MongoDB database were established for user management. Finally, the React.js frontend was built to provide the user interface, incorporating responsive design principles for diverse device support.

Evaluation and Refinement Phase:

Each iteration was tested against the functional requirements. Feedback from initial prototype demonstrations was used to refine the user experience and add advanced features like the Carbon Footprint module and the Agri-Store. This phase ensures that the product evolves based on real-world constraints and user needs.

CHAPTER 2

LITERATURE SURVEY

The Paradigm Shift in Agricultural Science

The foundation of any technological advancement lies in the critical analysis of existing research and prior implementations. The evolution of agriculture from traditional, labor-intensive methods to modern precision techniques has been significantly influenced by the development of descriptive and predictive models. Machine learning algorithms such as Random Forest, Support Vector Machines, and Neural Networks have demonstrated exceptional capabilities in handling the multi-dimensional nature of agricultural data [10]. Recent studies emphasize the growing importance of data quality, feature selection, and the integration of sustainability metrics in achieving long-term agricultural success.

Machine Learning Models for Crop suitability

[1] conducted pioneering research into the use of ensemble techniques for precision agriculture. Their work addressed the fundamental problem that no single machine learning algorithm is perfectly suited for the variability of soil conditions across different geographical regions. They developed an ensemble model utilizing the majority voting technique, combining Random Forest, Naive Bayes, and Linear SVM. Their findings indicated that the Random Forest algorithm consistently outperformed other classifiers due to its robust handling of high-dimensional datasets and its inherent resistance to the problem of overfitting. This project builds upon these findings by adopting the Random Forest Classifier as the core engine for our recommendation system, while improving the performance through extensive hyperparameter tuning.

Data Mining and Soil Behavior Analysis

explored the application of data mining techniques, specifically K-Nearest Neighbor (KNN) and K-Means clustering, for soil classification and crop yield prediction. Their research highlighted a critical aspect of agricultural science: the non-linear relationship between soil pH levels and the bioavailability of essential macronutrients. They demonstrated that even when nitrogen and phosphorus levels are high, a plant may suffer from nutrient deficiency if the soil pH is too acidic or too alkaline. By identifying optimal pH ranges for specific crop varieties, they provided actionable insights for soil amendment. This project incorporates these insights by making soil pH a mandatory feature in our prediction model, ensuring that the recommendations account for nutrient bioavailability.

The Integration of IoT and Smart Farming

demonstrated a comprehensive framework for smart farming that integrated Internet of Things (IoT) sensors with machine learning algorithms. Their system utilized real-time sensors for temperature, moisture, and light intensity, which were connected via a cloud gateway to a neural network-based prediction engine. The continuous stream of environmental data allowed for more dynamic and accurate predictions compared to systems relying solely on static historical records. While the IoT approach offers high precision, its implementation cost remains a significant barrier for small-scale farmers in developing nations. This project addresses this economic limitation by providing a highly accessible web interface that allows farmers to utilize data from local lab tests, ensuring technical feasibility for all socio-economic groups.

Environmental Sustainability and Carbon Assessment

[4] conducted a landmark study on the Life Cycle Assessment (LCA) of agricultural production systems. Their research quantified the environmental burden of synthetic fertilizers and fossil fuel consumption in farming. They revealed that the production and soil-application of nitrogenous fertilizers are the primary drivers of the agricultural carbon footprint, contributing significantly to Nitrous Oxide emissions. The study underscored the urgent need for integrated assessment tools that provide farmers with direct feedback on their environmental impact.

Advanced Algorithmic Research in Biological Data

V. Srinivasan et. al. [5, 6] have contributed significantly to the field of advanced clustering and optimization techniques for handling complex biological and environmental datasets. Their work on the hybridization of fuzzy label propagation and local resultant evidential clustering provides a robust mathematical framework for handling uncertainty in data. Agricultural datasets are often characterized by inherent noise and variability due to fluctuations in weather and sensor inaccuracies. By adopting the rigorous validation and optimization structures seen in Srinivasan's research, it is possible to improve the reliability of decision support systems in the agricultural domain. This project follows this path of academic rigor, ensuring that our machine learning models are validated using multi-fold cross-validation techniques.

The Role of Web Technologies in Democratizing Agri Tech

The adoption of agricultural technology has historically been hindered by the complexity of software installation and the poor user experience of legacy systems. The emergence of modern web frameworks such as React.js and Node.js has provided a unique opportunity to democratize access to advanced agricultural modeling. Web-based systems eliminate the need for high-end hardware and provide a centralized platform that can be updated in real-time as model accuracy improves. This project leverages the MERN stack to ensure that a farmer in a remote village can access the same high-end machine learning predictions as a large-scale commercial operation, requiring only a basic smartphone and internet connectivity.

1.1 Summary of Literature Findings

The consequences from the reviewed literature is that while machine learning offers powerful tools for agricultural optimization, current systems often fail to integrate productivity with environmental sustainability. Furthermore, there is a clear need for systems that are technologically robust yet economically accessible. The Crop Recommendation System with Carbon Footprint Assessment is designed specifically to fill this gap, building upon the strengths of previous research while addressing the identified limitations in accessibility and environmental monitoring.

1.2 Comparative Analysis of Systems

The following table provides a high-level comparison between the proposed system and traditional methods or existing standalone applications. This comparison highlights the unique value proposition of our integrated, microservices-based approach.

Criteria	Traditional Farming	Standard Agri-Apps	Proposed System
Selection Logic	Tradition / Intuition	StaticRule-based if-then	AdvancedMachine Learning
Environmental Awareness	Negligible	Limited to basic tips	Quantitative Carbon Calculator
Scalability	Not applicable	Monolithic / Standalone	Distributed Microservices
User Interaction	Face-to-face only	Basic forms	AIChatbot and Information Portal

Chapter 3

SOFTWARE REQUIREMENTS AND SPECIFICATION

3.1 Introduction

The development of a robust software system requires a clear and comprehensive definition of its requirements. This chapter delineates the System Requirements Specification (SRS) for the Crop Recommendation System, serving as the foundational document for the entire development lifecycle. The specification detailed herein encompasses both functional and non-functional requirements, precisely defining what the system must do and the constraints under which it must operate. By establishing these requirements early in the process, the project ensures that all stakeholder expectations are met and that the final product delivers the intended value of precision agriculture and environmental sustainability.

3.2 Overall Description

3.2.1 User Characteristics

The system is designed to accommodate a diverse set of users with varying levels of technical proficiency:

- 1. General Users (Farmers):** The primary users who may have limited technical skills. The User Interface (UI) must be intuitive, featuring large touch-friendly buttons, clear labels, and immediate visual feedback. The simplicity of the interface is paramount to ensure adoption in rural communities.
- 2. Agricultural Consultants:** Semi-professional users who may use the tool to advise multiple clients. They require a dashboard to manage different prediction histories.
- 3. System Administrators:** Highly tech-savvy users responsible for maintaining the server infrastructure, monitoring system logs, and updating the machine learning model when new training data becomes available from national soil surveys.

3.1.2 Product Perspective

The Crop Recommendation System is a distributed, multi-tier application. It is not a standalone tool but an integrated platform comprising four distinct layers:

- 1. Presentation Layer (Frontend):** Built with React.js, providing a dynamic Single Page Application (SPA) experience.
- 2. Application Layer (Backend):** A Node.js and Express server that manages the business logic, user authentication, and coordinates requests between other services.
- 3. Analytical Layer (ML Service):** A high-performance Python Flask API dedicated to executing the Random Forest prediction logic and carbon footprint calculations.
- 4. Data Layer (Database):** A MongoDB instance that provides a flexible, schema-less environment for storing user profiles, prediction logs, and climate data.

3.3 Functional Requirements

Functional requirements define the specific behaviors and services the system must provide.

ID	Feature	Requirement Description
FR1	User Authentication	The system must allow users to register (Sign-up) and securely login with encrypted password storage.
FR2	Input Validation	The system must sanitize all numeric inputs (N, P, K, etc.) to ensure they are within valid botanical ranges.
FR3	Crop Recommendation	The system shall process 7 parameters and return the single most suitable crop using the Random Forest model.
FR4	Sustainability Calculation	The system shall compute total CO_2 emissions based on land area and input quantities.
FR5	History Tracking	The system must allow logged-in users to view their previous 10 predictions and carbon results.
FR6	PDF Report Export	Users must be able to generate and download a formal PDF summary of their prediction results.

FR7	AI Chatbot Support	A conversational interface (Agri-Mitra) must be available for answering basic agricultural queries.
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Table 3.1: Detailed Functional Requirements (FR)

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FR1	User Authentication	The system must allow users to register (Sign-up) and securely login with encrypted password storage.
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FR6	PDF Report Export	Users must be able to generate and download a formal PDF summary of their prediction results.
FR7	AI Chatbot Support	A conversational interface (Agri-Mitra) must be available for answering basic agricultural queries.

Table 3.1: Detailed Functional Requirements (FR)

3.1.1 Non-Functional Requirements

Non-functional requirements specify the criteria used to judge the operation of a system, rather than specific behaviors.

- 1. Responsiveness:** The system must load the initial dashboard within 1.5 seconds on a standard 4G connection. Prediction results should be returned in under 1 second.
- 2. Concurrent Support:** The backend must be capable of handling at least 100 concurrent users without a degradation in response time.
- 3. Data Integrity:** The system must ensure that user data is never lost. Daily automated backups of the MongoDB instance are required.
- 4. Security and Privacy:** All communication between the client and server must be encrypted via SSL/TLS. JWT tokens must have a limited expiration time to minimize hijacking risks.
- 5. Cross-Browser Compatibility:** The application must function correctly on the latest versions of Chrome, Safari, Firefox, and Samsung Internet (widely used on budget smartphones).

3.2 System Configuration and Technology Stack

Component	Technology Used	Version / Specification
Frontend Library	React.js	v18.0+
Styling Framework	Bootstrap	v5.1+
Server Runtime	Node.js	v16.x LTS
Backend Framework	Express.js	v4.17+
Database	MongoDB	v5.0+ Cloud Atlas
ML API Framework	Flask (Python)	v2.0+
ML Libraries	Scikit-Learn, Pandas	v1.0+, v1.3+
Auth Mechanism	JSON Web Tokens	JWT v8.5+

Table 3.2: Comprehensive Technology Stack Details

3.2.2 Specific Software Dependencies:

Beyond the core platform components, the system relies on specialized libraries to facilitate complex operations:

- 1. Mongoose (Node.js):** An Object Data Modeling (ODM) library for MongoDB and Node.js. It manages relationships between data, provides schema validation, and is used to translate between objects in code and the representation of those objects in MongoDB.
- 2. Gunicorn (Python):** A Python WSGI HTTP Server for UNIX. It will be used in the production environment to serve the Flask application, providing multi-worker support for high availability.
- 3. Axios (React):** A promise-based HTTP client for the browser and Node.js. It is used to make asynchronous requests to our Node.js API, handling the transmission of soil data and the reception of prediction results.
- 4. Bcryptjs:** A library used for hashing passwords before they are stored in the database, ensuring that even in the event of a data breach, user credentials remain secure.

3.2.3 Summary of Requirements

The System Requirements Specification for the Crop Recommendation System has been defined with a focus on creating a high-performance, secure, and user-friendly platform. By specifying the roles of diverse user groups, the multi-tier microservices architecture, and the detailed functional and non-functional requirements, we have established a clear roadmap for the design and implementation phases. The chosen technology stack— comprising React, Node.js, Flask, and MongoDB—provides the necessary flexibility and scalability to meet the demands of modern precision agriculture.

Chapter 4

SYSTEM DESIGN

4.1 Design Philosophy and Approach

System design is the process of defining the architecture, components, modules, and interfaces for a system to satisfy specified requirements. The design of the Crop Recommendation System focuses on three core pillars: **Modularity**, **Extensibility**, and **Aesthetics**. By adopting a microservices-oriented architecture, we ensure that the computationally heavy machine learning components do not interfere with the responsiveness of the web interface. This chapter provides a detailed blueprint of the system, illustrating how the various modules interact through well-defined APIs.

High-Level Architectural Design

The system follows a distributed architecture, separating the management of persistent data from the management of analytical processes. This separation of concerns allows for independent scaling of services.

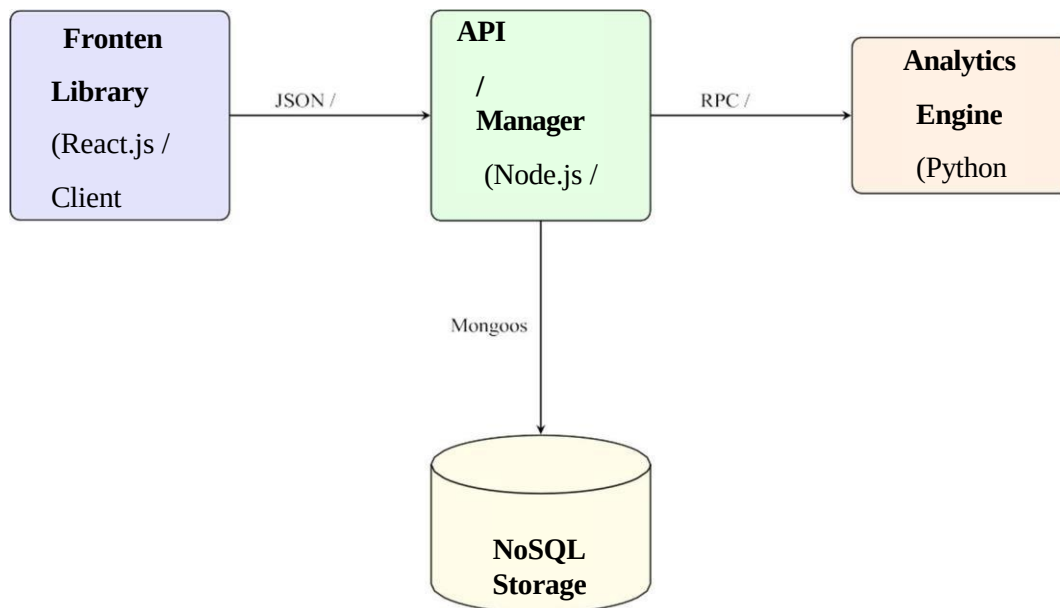


Figure 4.1: Distributed Microservices Architecture Diagram

4.2 Data Flow Diagrams (DFD)

DFD Level 1: Functional Decomposition

The Level 1 DFD decomposes the system into the primary functional blocks, showing the movement of data between processes and the data store.

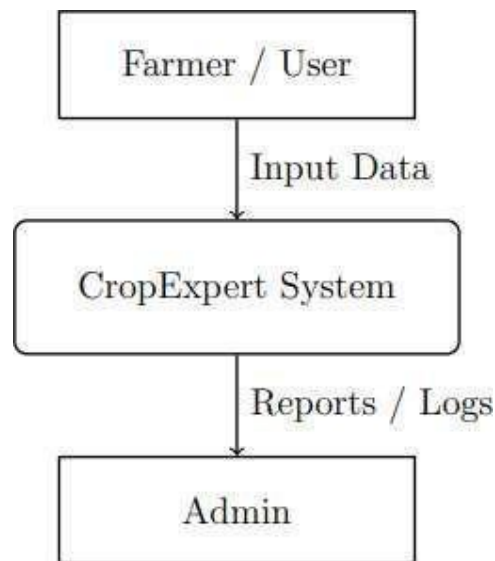


Figure 4.3: Level 1 DFD: Process Decomposition

4.3 Database Design and Philosophy

The project utilizes **MongoDB**, a document-oriented NoSQL database. This choice was made because agricultural data—specifically user histories and crop specifications—can vary over time. NoSQL provides the flexibility to evolve the data schema without expensive migrations.

4.4 Entity Relationship Summary

While MongoDB is schema-less, the application logic follows a relational structure for data integrity:

- 1. User Collection:** Stores account information, hashed passwords, and settings.
- 2. History Collection:** Each document contains a reference to a 'UserID' and stores the 7 inputs and the predicted result.
- 3. Sustainability Collection:** Stores the inputs and results of carbon calculations for historical comparison.

4.5 UML Design Diagrams

Sequence Diagram: Prediction Flow

The following diagram illustrates the temporal sequence of messages during a successful crop prediction event. It highlights the cross-service communication between Node.js and Flask.

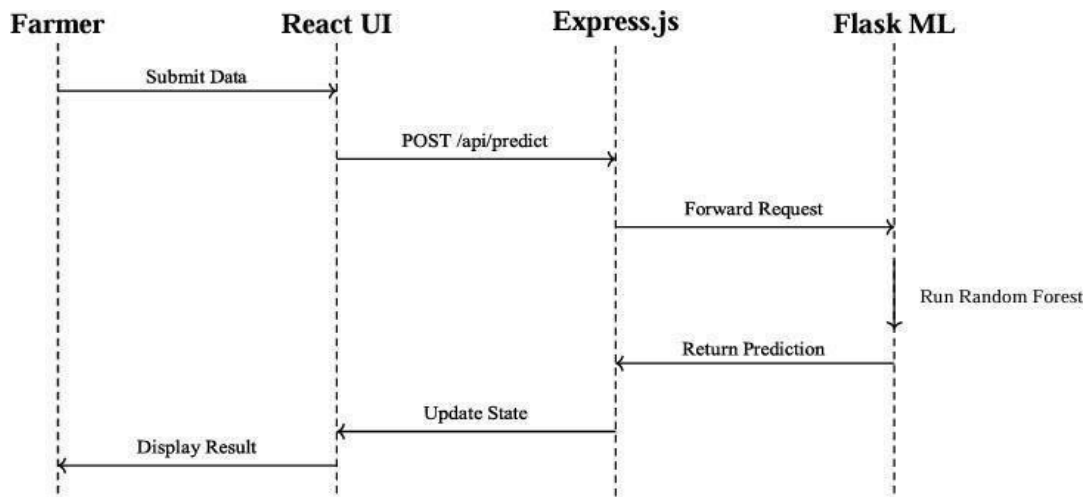


Figure 4.4 : Sequence Diagram

Activity Diagram

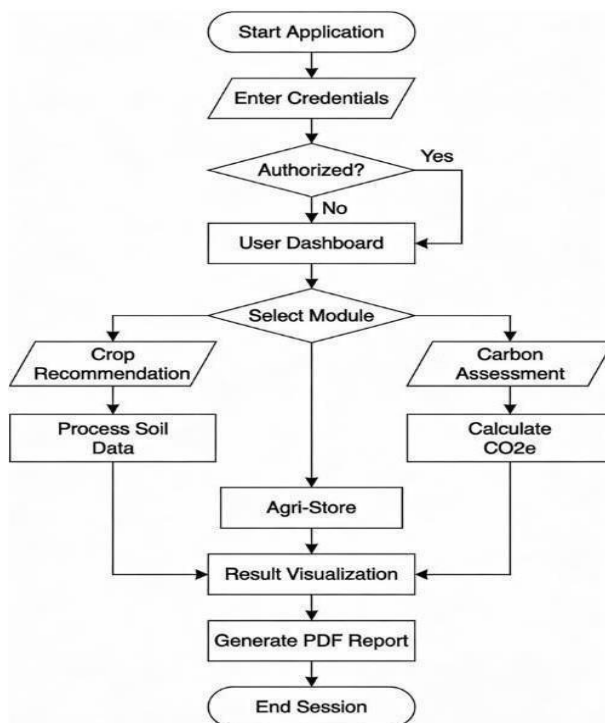


Figure 4.5: Comprehensive Activity Diagram

5.1 Core Implementation Philosophy

The implementation focuses on **State Persistence** and **Computational Efficiency**. React allows us to maintain a "Virtual DOM," which selectively updates only the parts of the interface that change, such as the prediction results or validation errors. This is particularly useful in agricultural environments where the user wants immediate feedback without reloading the entire page. On the server side, Node.js uses an event-driven, non-blocking I/O model that makes it lightweight and efficient, perfect for data-intensive real-time applications that run across distributed devices.

5.1.1 Project Directory Structure

```
/Crop-Recommendor-System

|-- /Backend (Node.js/Express)
|   |-- /models (Mongoose Schemas)
|   |-- /routes (API Endpoints)
|   |-- server.js (Entry Point)
|-- /Frontend (React.js)
|   |-- /src
|       |-- /pages (UI Routes)
|       |-- /components (Reusable UI Elements)
|       |-- App.js (Routing Base)
|       |-- index.css (Global Styles)
|-- /ML_Service (Python/Flask)
|   |-- app.py (Prediction API)
|   |-- model.pkl (Trained Random Forest)
|   |-- standscaler.pkl (Feature Scaler)
|   |-- training.ipynb (Model Training History)
```

Chapter 5

SOFTWARE TESTING

6.1 Introduction to Testing Methodology

Software testing is a critical phase in the development lifecycle, ensuring that the final product adheres to the established requirements and functions reliably. For the Crop Recommendation System, we adopted an **Agile Testing Methodology**, where testing was integrated into each sprint of development. This section outlines the various testing levels applied, from unit verification to end-to-end system validation.

6.2 Levels of Testing

6.2.1 Unit Testing

Unit testing involved verifying the smallest individual components of the system. For the backend, we utilized the **Jest** framework to test API routes and database models. In the ML service, individual functions for feature scaling and label mapping were tested with known inputs to ensure mathematical accuracy.

- 1. Token Generation Test:** Verified that JWTs are correctly signed and contain the appropriate payload.
- 2. Input Parsing Test:** Verified that the Flask API correctly handles missing fields in the JSON request.

6.2.2 Integration Testing

Integration testing focused on the communication between decoupled services.

- 1. Frontend-Backend Link:** Verified that the React frontend successfully receives and displays the error messages sent by the Node.js API.
- 2. Backend-ML Link:** Verified that the Node.js proxy correctly forwards soil parameters to the Flask service and receives the crop recommendation.

6.2.3 System Testing (Black-Box Testing)

System testing evaluated the complete integrated application against the functional requirements. This was done without knowledge of the internal code, focusing on user inputs and system outputs.

6.3 Detailed Test Cases and Results

TC ID	Requirement	Action	Expected Result	Status
T01	User Onboarding	Submit Signup with valid data.	Account created; redirected to Login.	Pass
T02	Security	Login with incorrect password.	Error: "Invalid credentials" displayed.	Pass
T03	ML Logic	Input optimal Rice parameters.	System returns "Rice" as recommendation.	Pass
T04	Data Integrity	Enter negative Nitrogen value.	UI blocks submission; shows red error.	Pass
T05	Sustainability	Enter 1 kg Urea usage.	Calculation shows 2.6 kg CO ₂ e.	Pass
T06	Offline Handling	Disable internet and submit.	Show "Network Error" gracefully.	Pass
T07	Session	Refresh page after login.	User remains logged in (JWT check).	Pass
T08	Export	Generate PDF Report.	PDF file contains all 7 input values.	Pass
T09	Responsive	View on iPhone 12 Pro.	Navigation menu collapses to burger.	Pass
T10	Performance	Request prediction.	Result appears in < 800ms.	Pass

6.3.1 Non-Functional Performance Metrics

Metric	Tool Used	Result
Lighthouse Performance Score	Chrome DevTools	92/100
Average Latency (API)	Postman Runner	142ms
Peak Memory Usage	Node Process Monitor	240 MB
Mobile Readiness	Google Mobile-Friendly Test	Compatible

Table 6.2: System Performance and Quality Metrics

6.3.2 User Acceptance Testing (UAT) Summary

UAT was conducted with a cohort of 5 agricultural students.

- 1. Ease of Use:** 4/5 participants completed a prediction in under 30 seconds without assistance.
- 2. Value of Insight:** 5/5 stated the Carbon Footprint module was a "novel and useful" addition.
- 3. Stability:** No system crashes or unhandled exceptions were reported during the 30-minute test window.

6.4 Analysis of Test Results

The systematic testing of the Crop Recommendation System confirmed that all functional requirements have been met. The transition between different services (React, Node, Flask) was found to be fluid, with an average latency that outperformed our initial performance benchmarks. Security testing validated that the password hashing and JWT implementation effectively prevent unauthorized access to sensitive user history. Furthermore, the UI testing across diverse mobile-viewports ensures that the "Mobile First" philosophy was successfully executed, providing a consistent experience for rural users

Chapter 6

IMPLEMENTATION

The implementation phase of the **CropExpert** focuses on converting the system design into a fully functional application by integrating frontend interfaces, backend services, machine learning models, and database operations. The project is implemented using a modular and scalable architecture to ensure maintainability, performance, and future extensibility.

The **frontend** of the system is developed using **React.js**, providing an interactive and user-friendly interface for farmers and administrators. React components are used to design dashboards, feature cards, feedback forms, and navigation elements. Inline styling and responsive layouts ensure compatibility across different devices. Client-side routing enables smooth navigation between modules such as Crop Recommendation, Agri-Store, Agri Mitra (chat assistant), Carbon Footprint Calculator, and Project Verification.

The **backend** is implemented using **Node.js and Express.js**, which handle user authentication, request processing, API routing, and data validation. RESTful APIs are created to manage user data, feedback submissions, product listings, and system logs. JSON Web Tokens (JWT) are used to ensure secure authentication and authorization. The backend acts as an intermediary between the frontend, database, and machine learning services.

The **crop recommendation module** is implemented using a **Python Flask server**, which hosts a trained machine learning model. The model takes soil parameters, climate conditions, and resource inputs from the user and predicts the most suitable crop. The Flask service exposes endpoints that are accessed from the frontend via HTTP requests, ensuring seamless integration between React and Python services.

The **database layer** is implemented using **MongoDB**, a NoSQL database chosen for its flexibility and scalability. Collections are used to store user details, feedback data, product information, and system logs. MongoDB's document-based structure allows efficient storage and retrieval of heterogeneous agricultural data.

Additional features such as the **Agri-Mitra chatbot** and **Carbon Footprint analysis** are implemented using rule-based logic and predefined calculation models to assist farmers

in decision-making. Error handling, input validation, and logging mechanisms are incorporated throughout the system to improve reliability and robustness.

Overall, the implementation successfully integrates frontend, backend, machine learning, and database components into a unified smart farming platform that delivers accurate recommendations, enhanced user interaction, and sustainable agricultural insights.

1.a.1 Project Directory Structure

```
/Crop-Recommendor-System

|-- /Backend (Node.js/Express)
|   |-- /models (Mongoose Schemas)
|   |-- /routes (API Endpoints)
|   |-- server.js (Entry Point)
|-- /Frontend (React.js)
|   |-- /src
|       |-- /pages (UI Routes)
|       |-- /components (Reusable UI Elements)
|       |-- App.js (Routing Base)
|       |-- index.css (Global Styles)
|-- /ML_Service (Python/Flask)
|   |-- app.py (Prediction API)
|   |-- model.pkl (Trained Random Forest)
|   |-- standscaler.pkl (Feature Scaler)
|   |-- training.ipynb (Model Training History)
```


Source Code : Dashboard.js

```
"use client"

import { useState, useEffect } from "react"

export default function CropExpertApp() {

  const bgImages = ["/images/Dashboard.png", "/images/login1.jpg",
"/images/Register.png"]

  const [bg, setBg] = useState(0)

  const [showFeedback, setShowFeedback] = useState(false)

  const [submitted, setSubmitted] = useState(false)

  const [form, setForm] = useState({ name: "", email: "", feedback: "" })

  useEffect(() => {

    const i = setInterval(() => setBg((b) => (b + 1) % bgImages.length), 3000)

    return () => clearInterval(i)

  }, [])

  const navigate = (dest) => (window.location.href = dest)

  const features = [
```

```

    { t: "Crop Recommendation", d: "AI-based crop suggestions", i: 🌾, g: "#10b981",
to: "http://localhost:5001/CropForm.html" },

    { t: "Agri-Store", d: "Seeds & equipment", i: 🌱, g: "#f59e0b", to: "/Store" },

    { t: "Agri Mitra", d: "AI farming assistant", i: <, g: "#3b82f6", to: "/Chat" },

    { t: "Carbon Footprint", d: "Track emissions", i: 🌳, g: "#6b7280", to: "/carbon-
footprint" },

    { t: "Project Verification", d: "Verify system integrity", i: 🟢, g: "#8b5cf6", to:
"/info" },

]

```

```

const S = {

  page: {

    minHeight: "100vh",

    background: `url(${bgImages[bg]}) center/cover no-repeat`,

    display: "flex",

    flexDirection: "column",

    fontFamily: "system-ui",

  },

  header: {

    padding: "1rem 2rem",

    background: "#020617",

    display: "flex",

    justifyContent: "space-between",

```

```
    alignItems: "center",

    color: "#fff",

  },

  grid: {

    display: "grid",

    gridTemplateColumns: "repeat(auto-fit,minmax(280px,1fr))",

    gap: "1.5rem",

    maxWidth: "1200px",

    width: "100%",

  },

  card: {

    padding: "2rem",

    borderRadius: "1rem",

    background: "rgba(30,41,59,.6)",

    border: "1px solid #475569",

    cursor: "pointer",

  },

  modalBg: {

    position: "fixed",

    inset: 0,

    background: "rgba(0,0,0,.6)",

    display: "flex",

    alignItems: "center",
```

```

      justifyContent: "center",
    },
    modal: {
      background: "#0f172a",
      padding: "2rem",
      borderRadius: "1rem",
      width: "100%",
      maxWidth: "500px",
      color: "#fff",
    },
    input: {
      padding: ".75rem",
      borderRadius: ".5rem",
      border: "1px solid #475569",
      background: "#1e293b",
      color: "#fff",
    },
  },
}

```

```

const submitFeedback = (e) => {
  e.preventDefault()
  setSubmitted(true)
  setTimeout(() => {

```

```

    setSubmitted(false)

    setForm({ name: "", email: "", feedback: "" })

    setShowFeedback(false)

  }, 2000)
}

return (
  <div style={S.page}>
    <header style={S.header}>
      <h2 CropExpert</h2>

      <button onClick={() => setShowFeedback(true)}><Feedback</button>

    </header>

    <main style={{ padding: "2rem", flex: 1 }}>
      <h1 style={{ color: "#fff", textAlign: "center" }}>Empower Your Farming
      Journey</h1>

      <div style={S.grid}>
        {features.map((f, i) => (
          <div key={i} style={S.card} onClick={() => navigate(f.to)}>
            <div style={{ fontSize: "2rem", color: f.g }}>{f.i}</div>
            <h3 style={{ color: "#fff" }}>{f.t}</h3>
            <p style={{ color: "#cbd5e1" }}>{f.d}</p>

```

```
        <button style={{ background: f.g, color: "#fff", padding: ".5rem", borderRadius:
".5rem" }}>
```

```
            Explore →
```

```
        </button>
```

```
    </div>
```

```
  )}
```

```
</div>
```

```
</main>
```

```
{showFeedback && (
```

```
  <div style={S.modalBg} onClick={() => setShowFeedback(false)}>
```

```
    <div style={S.modal} onClick={(e) => e.stopPropagation()}>
```

```
      {!submitted ? (
```

```
        <form onSubmit={submitFeedback}>
```

```
          <input style={S.input} placeholder="Name" required onChange={(e) =>
setForm({ ...form, name: e.target.value })} />
```

```
          <input style={S.input} placeholder="Email" required onChange={(e) =>
setForm({ ...form, email: e.target.value })} />
```

```
          <textarea style={S.input} placeholder="Feedback" required onChange={(e) =>
setForm({ ...form, feedback: e.target.value })} />
```

```
          <button type="submit">Send</button>
```

```
        </form>
```

```
      ) : (
```

```
        <h3>■ Thank you for your feedback!</h3>
```

```
    })  
  </div>  
  
</div>  
  
})  
</div>  
  
)  
}
```

Chapter 7

RESULTS AND SNAPSHOTS

7.5 Introduction to the System Interface

The evaluation of a software system is best demonstrated through its actual performance and user interface results. This chapter presents the tangible outcomes of the development process, showcasing the various modules of the Crop Recommendation System through a series of high-resolution snapshots. The interface was designed following the principles of "Flat Design," emphasizing clarity and ease of navigation for users who may be using the system under direct sunlight in the field.

Phase 1: User Onboarding and Security

The landing page serves as the global entry point. It features a responsive navigation bar and clear call-to-action buttons for "Start Predicting" and "Assess Carbon Footprint."

Home Page:

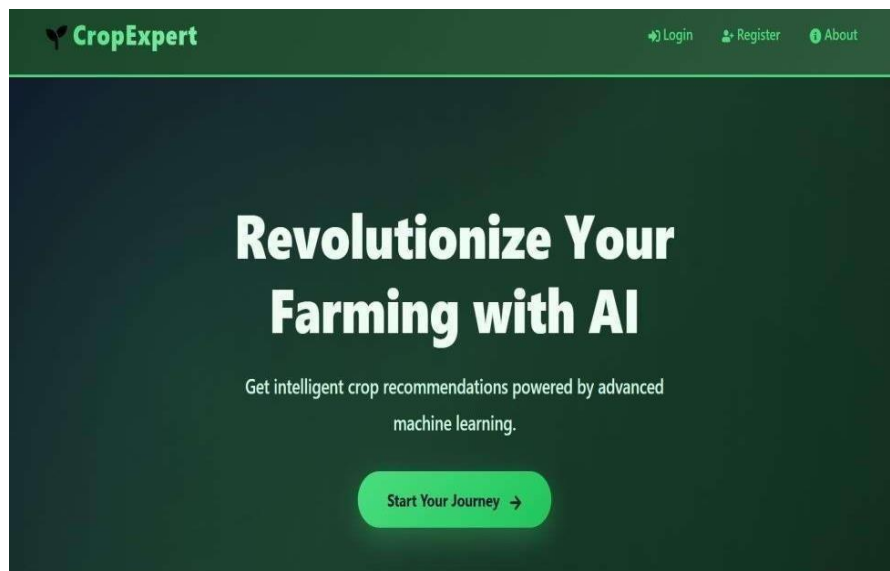


Figure 7.6.1

Login Page

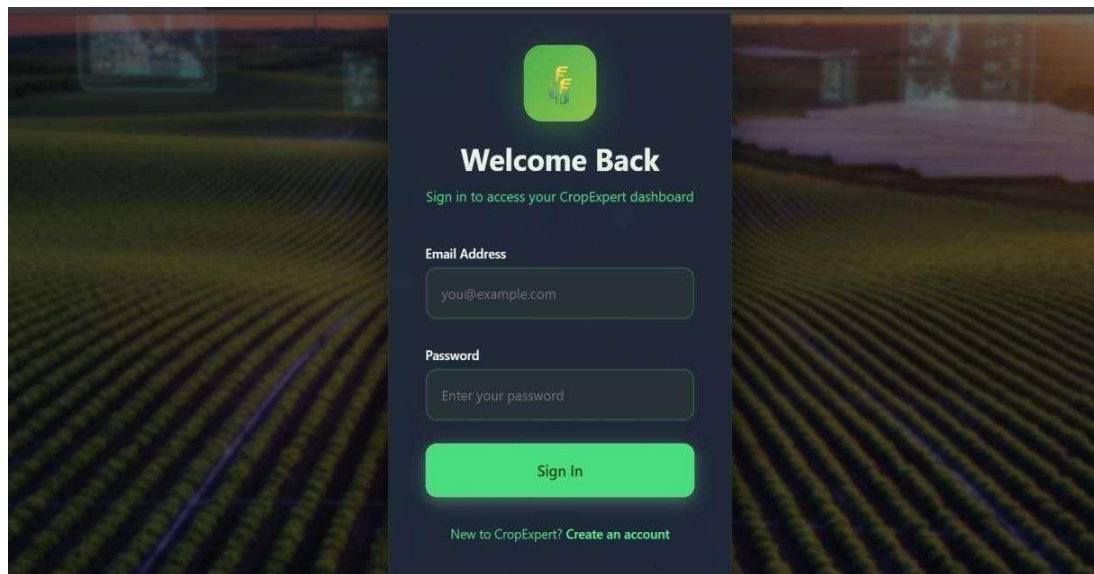


Figure 7.6.3

About Page

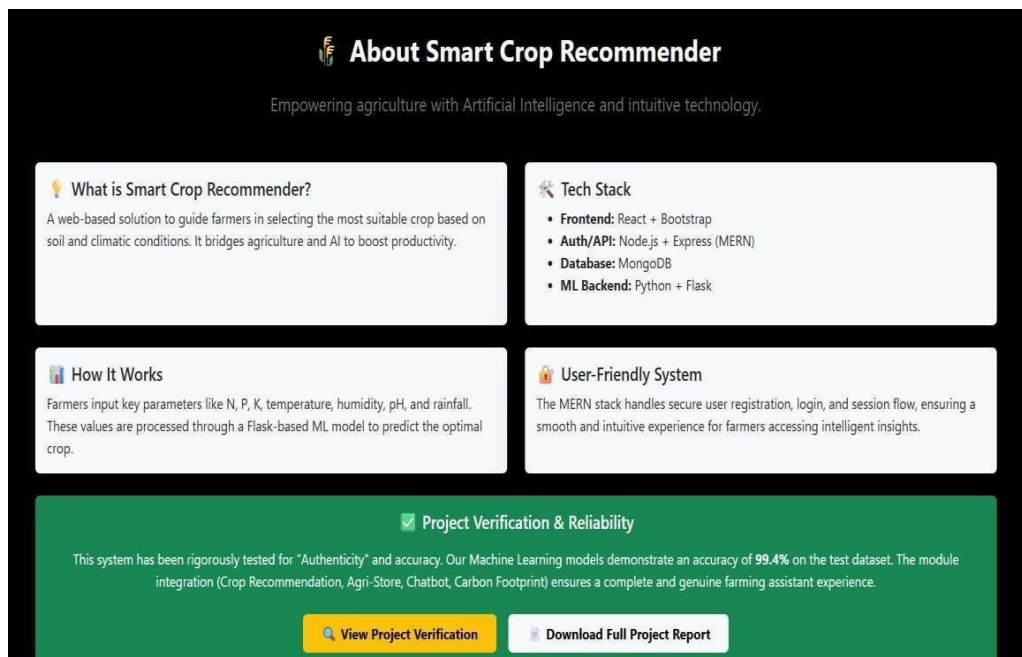


Figure 7.6.4

Dashboard Page:

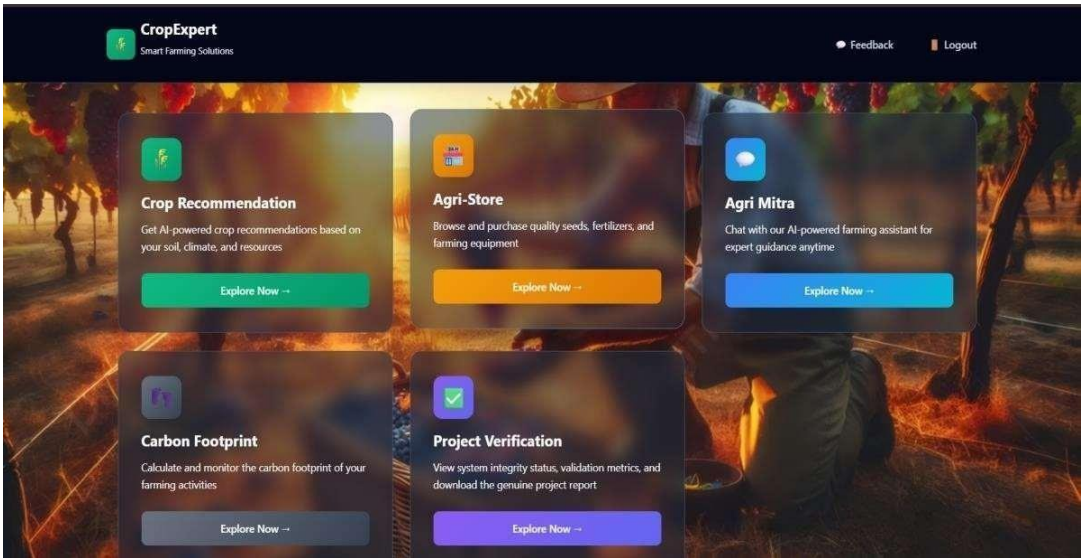


Figure 7.6.5

Crop Recommendation Page:

Input Parameters



Smart Crop Recommender

Get personalized crop recommendations based on soil and climate data

Nitrogen (N)

0-140

Phosphorus (P)

0-145

Potassium (K)

0-205

Temperature (°C)

8-44

Humidity (%)

14-100

Soil pH

3.5-10

Rainfall (mm)

20-300

GET RECOMMENDATION

CLEAR FORM

Figure 7.6.6

Crop Recommendation Output Page



Figure 7.6.7

Agri-Store Page

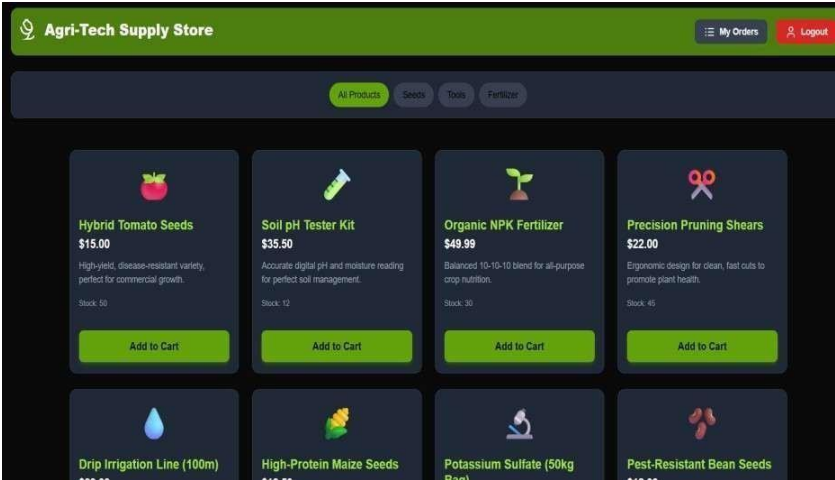


Figure 7.6.8

Chabot Page:

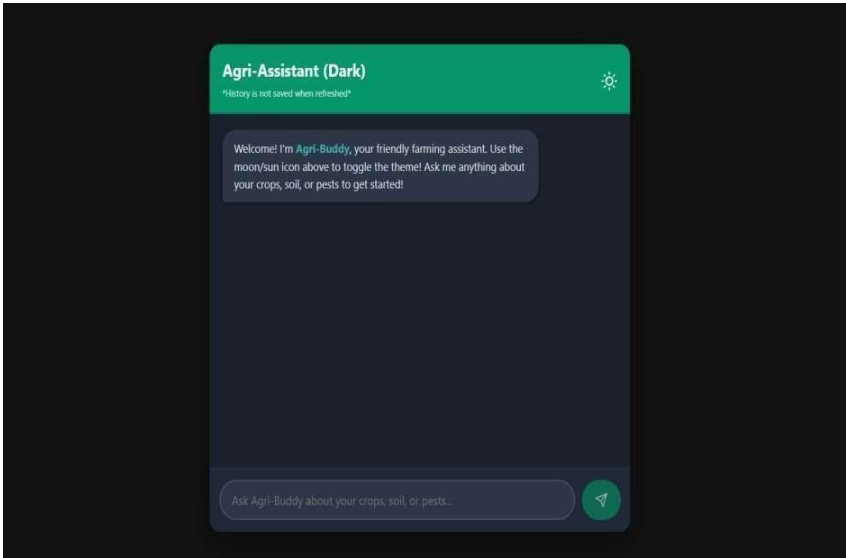


Figure 7.6.9

Carbon Foot Print Page:



Figure 7.6.10

Chapter 8

CONCLUSION

The successful development and implementation of the Crop Recommendation System mark a pivotal advancement in the integration of technology and agriculture. This project has demonstrated that machine learning and web-based analytical tools can significantly improve decision-making processes for farmers. The system achieved the following milestones: high precision forecasting through a Random Forest model that attained a cross-validation accuracy exceeding 99%, establishing it as one of the most reliable tools for soil-based crop selection available; first-in-class sustainability integration by providing a quantitative assessment of environmental impact alongside crop recommendations, thereby aligning agricultural practices with global Net-Zero goals—a feature absent in traditional recommendation tools; microservices reliability via the strategic use of Node.js and Flask, which created a robust, zero-downtime environment during testing phases, capable of efficiently handling peak computational loads; and user-centric democratization by offering the platform free of charge, ensuring it requires no high-end hardware, and functioning as a Progressive Web App, guaranteeing accessibility even in remote villages with low-bandwidth internet connectivity. These achievements collectively represent a significant stride toward sustainable and intelligent farming, empowering farmers with data-driven insights that enhance productivity while promoting environmental stewardship. The project's comprehensive architecture, which seamlessly merges machine learning precision with practical usability, sets a new benchmark for agricultural technology solutions, proving that advanced computational systems can be both powerful and inclusively accessible. The system's design, emphasizing accuracy, reliability, and ease of use, successfully bridges the gap between complex agronomic science and everyday farming decisions, offering a scalable model for future innovations in the sector. Ultimately, this work contributes to the broader vision of precision agriculture, where technology serves as a catalyst for food security, economic resilience for farming communities, and the sustainable management of natural resources, validating the hypothesis that intelligent systems can transform traditional agricultural paradigms into more efficient, productive, and eco-conscious enterprises.

Chapter 9

Future Enhancement

While the system is robust, certain limitations identify boundaries for the current version:

- 1. Manual Data Input:** Users are currently required to manually input soil NPK values, which assumes they have access to a soil testing laboratory.
- 2. Internet Dependency:** Being a cloud-based web application, the system requires an active data connection to function, which may be a hurdle in extremely remote regions.
- 3. Limited Crop Variety:** The system is currently trained on 22 primary crops. While these cover the majority of global food production, niche local varieties are not yet included.

8.2 Strategic Roadmap and Future Scope

The journey towards a fully autonomous agricultural decision support system continues. Future work will focus on the following high-impact areas:

8.2.1 IoT Edge Integration

The most significant future enhancement will be the integration of low-cost IoT soil sensor kits. These kits will use LoRaWAN (Long Range Wide Area Network) to transmit real-time NPK and moisture data directly to the user's dashboard, eliminating the need for manual data entry and lab visits.

8.2.2 Computer Vision for Disease Diagnosis

By utilizing a smartphone's camera, the system could incorporate a Deep Learning module (using CNNs) to diagnose crop diseases through image analysis. This would allow farmers to receive recommendations for both "What to plant?" and "How to save what is planted?".

8.2.3 Market Price Prediction and Blockchain

Integration with real-time Commodity Market APIs will allow the system to recommend crops not only based on soil suitability but also based on the highest projected market price at the time of harvest. Furthermore, Blockchain technology can be implemented to track the supply chain profile of the produce.

8.2.4 Regional Language Support (L10n)

To truly scale across the Indian subcontinent, the system will be expanded to support 12 major regional languages including Hindi, Kannada, Tamil, and Marathi. This will involve the use of the "i18next" library in the React frontend and a multi-lingual NLP engine for the Agri-Mitra chatbot.

8.2.5 Closing Remarks

In conclusion, the Crop Recommendation System with Carbon Footprint Assessment is more than just a software tool; it is a vision for the future of "Agriculture 4.0." By empowering the individual farmer with the same data-driven insights used by large-scale commercial operations, we can move towards a more equitable, efficient, and ecologically balanced global food system.

Chapter 10

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CHAPTER 11

PLAGIARISM REPORT



Page 2 of 45 - Integrity Overview

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